

MicroSAL — Backend Map & Native Support

Color	Meaning
Green	Full-featured / desktop OS
Blue	Real-time kernel (RTOS)
Orange	Minimal / cooperative

POSIX-Family

POSIX
(any POSIX:2008)

QNX
(Neutrino RTOS)

NuttX
(Apache, ≥ 12)

Linux
(≥ 4.11, extends POSIX)

RTEMS
(POSIX profile)

INTEGRITY
(pthread profile)

Shared implementation files in ``src/common/posix/``:

- ``posix_clock.inl`` — all 6 backends
- ``posix_rwlock.inl`` — all 6 backends
- ``posix_condvar.inl`` — POSIX, Linux, QNX, RTEMS, INTEGRITY
- ``posix_pthread_work_queue.inl`` — POSIX, Linux, RTEMS, INTEGRITY

Tick-width guard:
``OSAL_CFG_NATIVE_TICK_BITS``
enforces ``sizeof(osal::tick_t)``
≤ native tick width at compile time.

Native condvar: ✓ (pthread_cond_t, CLOCK_MONOTONIC)
Native rwlock: ✓ (pthread_rwlock_t)
Native work queue: ✗ (emulated)
Wait-set: ✓ (epoll)
Timed join: ✗

Commercial RTOS

ThreadX
(Azure RTOS)

PX5
(ThreadX superset)

FreeRTOS
(all chip families)

Zephyr
(≥ 3.0)

VxWorks
(Wind River 7)

Native condvar: ✓ (k_condvar)
Native event flags: ✓ (k_event)
Native work queue: ✓ (k_work_queue)
Native memory pool: ✗ (emulated)
ISR semaphore: ✓
Timed join: ✓

Native condvar: ✗ (emulated)
Native work queue: ✗ (emulated)
ISR semaphore: ✓
Event flags: ✓ (native xEventGroup)
Stream buffer: ✓ (xStreamBufferCreate)
Message buffer: ✓ (xMessageBufferCreate)
ISR stream buffer: ✓

ARM CMSIS

CMSIS-RTOS
(v1, cmsis_os.h)

CMSIS-RTOS2
(v2, cmsis_os2.h)

Native event flags: ✓ (osEventFlags)
Native memory pool: ✓ (osMemoryPool)
ISR semaphore: ✓
Recursive mutex: ✓

Native memory pool: ✓ (osPool)
ISR semaphore: ✓
Event flags: emulated (osSignal is per-thread)

No-OS

Bare-metal
(cooperative, single-core)

Cooperative round-robin.
No preemption. Priority ignored. All advanced features emulated.